

Introduction to Machine Learning

Mid-Term Examination — Syllabus

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Scope

1. NumPy — Arrays and Indexing
2. NumPy — Operations and Reductions
3. Pandas — Selecting Data
4. Pandas — Data Cleaning
5. Pandas — Grouping and Aggregation
6. Pandas — Saving Data
7. Data Visualisation
8. scikit-learn — Splitting, Scaling and Accuracy
9. Machine Learning Concepts

Every function that can appear in this paper is named below. **If it is not on this list, it will not be asked.**

NumPy — Arrays and Indexing

CREATING ARRAYS

Making an array from a list, or generating one of a fixed shape or range.

`np.array()` · `np.zeros()` · `np.ones()` · `np.full()` · `np.eye()` · `np.arange()` · `np.linspace()`

ATTRIBUTES AND TYPES

Checking the shape, size and data type of an array, and converting the type.

`shape` · `ndim` · `size` · `dtype` · `astype()`

INDEXING AND SLICING

Picking single values, a row, a column, or a block of both.

`a[i]` · `a[i, j]` · `a[1:3]` · `a[:, 0]` · `a[1, :]` · `a[1:3, 0:2]`

BOOLEAN AND FANCY INDEXING

Picking values that satisfy a condition, or a chosen set of positions.

`a[a > 5]` · `&` · `|` · `a[[0, 2, 4]]` · assigning values through a mask

NumPy — Operations and Reductions

ELEMENT-WISE OPERATIONS

Applying an operation to every value at once, without a loop.

`+` · `-` · `*` · `/` · `**` · comparison operators · `np.sqrt()` · `np.round()`

BROADCASTING

Combining arrays of different shapes, and knowing when the shapes do not fit.

array with a scalar · 2-D array with a 1-D row · identifying a shape mismatch

RESHAPING AND JOINING

Changing the shape of an array, and stacking arrays together.

`reshape()` · `reshape(-1, n)` · `ravel()` · `flatten()` · `.T` · `np.hstack()` · `np.vstack()` · `np.concatenate()`

REDUCTIONS

Collapsing values into a single number, or one value per row or column.

`sum()` · `mean()` · `min()` · `max()` · `std()` · `var()` · `argmin()` · `argmax()` · `np.sort()` · `np.unique()` · `axis=0` · `axis=1`

RANDOM NUMBERS

Generating random values, and fixing them so the result repeats.

`np.random.rand()` · `np.random.randn()` · `np.random.randint()` · `np.random.seed()`

Pandas — Selecting Data

LOADING AND FIRST LOOK

Reading a CSV and checking its size, columns, types and basic statistics.

`pd.read_csv()` · `head()` · `tail()` · `sample()` · `shape` · `columns` · `dtypes` · `info()` · `describe()`

SELECTING COLUMNS

Taking one column or several columns out of a DataFrame.

`df['col']` · `df[['a', 'b']]`

ILOC AND LOC

Selecting rows and columns by position, or by label.

`df.iloc[rows]` · `df.iloc[rows, cols]` · `df.loc[rows]` · `df.loc[rows, cols]` · `df.loc[5:10, 'total_bill':'time']` · `df.iloc[mask.values]`

FILTERING ROWS

Keeping only the rows that satisfy one or more conditions.

```
df[df['Age'] > 30] · & · | · ~ · df.loc[condition, 'col']
```

INSPECTING AND CHANGING VALUES

Counting categories, listing unique values, and updating the values that match a condition.

```
value_counts() · unique() · nunique() · df.loc[condition, 'col'] = value
```

SHORTCUTS

Matching against a set of values, checking a range, sorting, and handling the index.

```
isin() · between() · sort_values() · ascending= · reset_index() · set_index()
```

Pandas — Data Cleaning

MISSING VALUES

Finding blanks, counting them per column, and either removing or filling them.

```
isnull() · notnull() · isnull().sum() · dropna() · subset= · axis= · how= · fillna() · mean() · median() · mode()
```

DUPLICATES

Finding and removing repeated rows.

```
duplicated() · drop_duplicates()
```

COLUMNS AND TYPES

Dropping a column, and converting a column to the correct type.

```
drop(columns=) · astype()
```

Pandas — Grouping and Aggregation

GROUPING

Splitting the rows by one category and calculating a value for each group.

```
groupby('col')
```

AGGREGATING

Calculating a value for each group, and naming the resulting column.

```
mean() · sum() · count() · min() · max() · agg() · named aggregation · size() · size() vs count()
```

Pandas — Saving Data

WRITING TO A FILE

Saving a DataFrame back out as a CSV file.

```
to_csv() · index=False
```

Data Visualisation

PLOTS

Choosing the plot that answers the question being asked.

```
sns.histplot() · sns.countplot() · sns.boxplot() · sns.scatterplot() · sns.pairplot() · sns.heatmap() · df.corr()
```

ARGUMENTS

Setting what goes on each axis, which DataFrame to use, and splitting by a third column.

```
x= · y= · data= · hue= · bins= · kde= · annot= · cmap=
```

LABELLING

Setting the figure size, the title and the axis labels.

```
plt.figure(figsize=) · plt.title() · plt.xlabel() · plt.ylabel() · plt.show()
```

READING A PLOT

Saying what the shape of a plot tells you about the data.

choosing the right plot for a given question · a skewed histogram · box width and outliers · a pattern in a scatterplot · positive, negative or near-zero correlation

scikit-learn — Splitting, Scaling and Accuracy

Only the three functions below are in this exam. No model is to be built.

SPLITTING THE DATA

Keeping part of the data aside so the model can be tested on rows it has not seen.

```
train_test_split() · test_size= · random_state= · X_train · X_test · y_train · y_test
```

SCALING

Bringing columns onto a comparable range, learnt from the training data only.

```
StandardScaler() · fit_transform() on the training data · transform() on the test data
```

ACCURACY

Comparing predictions against the true values.

```
accuracy_score(y_test, y_pred)
```

IMPORTS

Where each of these comes from.

```
from sklearn.model_selection import train_test_split · from sklearn.preprocessing import StandardScaler · from sklearn.metrics import accuracy_score
```

Machine Learning Concepts

This section is **theory only**. Each point below should be explained in two or three sentences in your own words. No code is required here.

WHAT MACHINE LEARNING IS

- **How learning from data differs from writing the rules yourself**
Where the rule comes from in each case.
- **Identifying the features and the target in a given dataset**
Given a table of columns, which column is predicted and which are used to predict it.
- **Where machine learning fits, and where a simple rule would do the job better**
One example of each.

TYPES OF PROBLEMS

- **Supervised vs unsupervised learning**
What decides which one a dataset belongs to.
- **Classification vs regression**
What decides which one a problem is, with one example of each.
- **Placing a given problem statement into the right category**
The reason carries as much weight as the label.

HOW A MODEL IS BUILT

- **What scikit-learn is, and why a common library is used**
What it provides.
- **The main module groups of scikit-learn**
Preprocessing, model selection, models, metrics — one line on each.
- **The common pattern every scikit-learn tool follows**
The order in which a tool is created, learns, and is then used.
- **The stages of building a model, from raw data to a final score**
The full sequence, in order, from memory.
- **What each stage does, and why the order cannot be changed**
One line per stage, and what breaks if two are swapped.

JUDGING A MODEL

- **Why a model cannot be judged on the data it learnt from**
What the score would actually be measuring.
- **Why columns on very different ranges need to be scaled**
What goes wrong without it.
- **What an accuracy score means**
What it compares, and how a value is read.
- **What R-squared tells you about a model**
When it is used, and what a high or low value indicates.

Not Included in the Mid-Term

These topics are part of the course and will be examined later in the semester. Do not spend your preparation time on them now.

The KNN algorithm and `KNeighborsClassifier` · choosing a value of `k` · encoding categorical columns · `SimpleImputer` · `ColumnTransformer` · Linear Regression · Logistic Regression · calculating RMSE or R-squared

Datasets Used in Class

Questions will be based on these datasets or on data with a similar structure.

Dataset	Columns
<code>tips.csv</code>	total_bill, tip, sex, smoker, day, time, size
<code>Titanic-Dataset.csv</code>	Survived, Pclass, Sex, Age, SibSp, Parch, Fare, Cabin, Embarked
<code>planets.csv</code>	method, number, orbital_period, mass, distance, year
<code>mpg.csv</code>	mpg, cylinders, displacement, horsepower, weight, acceleration, model_year, origin, name

How to Prepare

- Re-run the class notebooks and make sure you can explain why each output looks the way it does.
- Before running a cell, say what you expect the output to be. A large part of this paper is reading code and stating what it produces.

- Type the selection, filtering and grouping code yourself rather than reading it.
- For the concepts section, be able to explain each point in two or three sentences in your own words.
- Work from this list. Every function you need is named here.